

Bounded validation report

System: pt3_gainsnzn_bounded_validation

Status: passed

Validated scope: Software configuration, interpolation inside the fixed L12-Pt3(Ga,In,Sn,Zn) representative-occupancy manifold, and rank consistency on the bundled M-Ga binary reference set.

software_configurability: demonstrated_for_bundled_inputs

within_manifold_interpolation: validated_internal

binary_energy_rank_consistency: observed_reference_set

new_host_or_prototype_transferability: not_evaluated

nonmetal_compound_transferability: not_evaluated

synthesizability: not_predicted

This report does not predict synthesizability.

Fixed-manifold validation

Module: fixed_l12_manifold

Composition rows: 165

Mixing elements: Ga, In, Sn, Zn

Pair parameters: 6

Training R2: 0.999277342

Training RMSE: 0.135821 kJ mol⁻¹ atom⁻¹

Endmember-only RMSE: 1.136521

Non-endmember LOOCV RMSE: 0.142250

Group-holdout RMSE range: 0.071806-0.231962

Interpretation: internal interpolation and ablation only.

Energy-backend reference comparison

Module: uma_mp_dft_binary_rank

Backends: Materials Project DFT, UMA-s-1p1, CHGNet

Unit: kJ mol⁻¹ atom⁻¹

all_hosts mp_dft__uma_s_1p1: N=15, rho=0.8679, RMSE=10.484, top-5 overlap=4, reversals=15/105

all_hosts mp_dft__chgnet: N=15, rho=0.9500, RMSE=6.063, top-5 overlap=5, reversals=8/105

all_hosts uma_s_1p1__chgnet: N=15, rho=0.9036, RMSE=8.841, top-5 overlap=4, reversals=13/105

size_compatible_hosts mp_dft__uma_s_1p1: N=12, rho=0.9510, RMSE=5.966, top-5 overlap=4, reversals=5/66

size_compatible_hosts mp_dft__chgnet: N=12, rho=0.9371, RMSE=5.444, top-5 overlap=5, reversals=5/66

size_compatible_hosts uma_s_1p1__chgnet: N=12, rho=0.9720, RMSE=7.708, top-5 overlap=4, reversals=4/66

Limitation: This is a heterogeneous binary reference-set consistency check, not a direct accuracy validation of the five-component energy landscape or a synthesis-success test.